

Cosmic Universalism Website

Local Website Implementation Manual

A controlled, step-by-step workflow for building a local Astro website with ChatGPT Desktop, PyCharm, GitHub Desktop, and GitHub Pages.

Repository	CosmicUniversalismStatement
Development branch	website
Website directory	website/
Primary framework	Astro with TypeScript
Editor	PyCharm with the Astro plugin
Version control	GitHub Desktop
Planning environment	ChatGPT Work
Implementation environment	ChatGPT Codex
Document version	1.0
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<https://github.com/willmaddock/CosmicUniversalismStatement>

Document Control

Document	Cosmic Universalism Website: Local Website Implementation Manual
Version	1.0
Status	Initial controlled implementation guide
Repository	CosmicUniversalismStatement
Protected branch	main
Development branch	website
Website source directory	website/
Intended platform	macOS with GitHub Desktop and PyCharm
Deployment target	GitHub Pages
Last technical verification	July 18, 2026

Software interfaces and action versions can change. Before the final deployment, compare the deployment section with the current official Astro and GitHub documentation. The local build and branch-protection principles remain valid even when interface labels change.

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Final Operating Principle

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1 Purpose and Intended Outcome

Purpose

This manual guides the creation of a local, production-quality website for the **Cosmic Universalism Statement** repository while protecting the existing research, converter, media, and main branch.

The intended repository arrangement is:

Target repository structure

```
CosmicUniversalismStatement/  
|-- .github/  
|-- ResearchFiles/  
|-- cosmic_converter/  
|-- Images/  
|-- media/  
|-- website/  
|   |-- public/  
|   |-- src/  
|   |-- astro.config.mjs  
|   |-- package.json  
|   |-- package-lock.json  
|   |-- tsconfig.json  
|-- README.md  
|-- LICENSE.md
```

The branch arrangement is:

Branch strategy

```
main  
`-- Canonical Cosmic Universalism research repository  
  
website  
`-- Website source, website content, and website deployment workflow
```

Safety Rule

The website branch is a separate version of the complete repository. Work performed on that branch does not alter main unless changes are deliberately merged into main.

2 Application Responsibilities

Application	Primary responsibility
GitHub Desktop	Create and switch branches, review file changes, create commits, fetch, push, and publish branches.
PyCharm	Inspect and edit Astro, TypeScript, HTML, CSS, Markdown, SVG, and configuration files. Run the local terminal and inspect project files.
ChatGPT Work	Plan phases, structure research, write specifications, organize content, review design decisions, and prepare implementation tasks.
ChatGPT Codex	Inspect the repository, create or edit code, run commands, execute builds, report errors, and provide file-by-file change summaries.
Browser	View the website at the local Astro development address and test desktop and mobile behavior.
Astro	Generate the static production website.
GitHub Pages	Host the generated public website after the local build is approved.

Checkpoint

Each application has one primary responsibility. Avoid asking several applications to manage Git simultaneously. Use GitHub Desktop as the main visual source-control interface.

3 Phase 1: Preflight Checklist

Complete this checklist before creating website files.

- ☐ GitHub Desktop is installed and signed in.
- ☐ The repository is available locally.
- ☐ PyCharm is installed.
- ☐ The JetBrains Astro plugin is installed.
- ☐ Node.js 22.12.0 or newer is installed.
- ☐ Node.js 24 LTS is preferred when available.
- ☐ npm is available.
- ☐ ChatGPT Desktop is installed and signed in.
- ☐ The approved homepage mockup is saved.
- ☐ The website blueprint is saved.

3.1 Verify Node.js and npm

Open Terminal or PyCharm's built-in terminal and run:

Verify Node.js and npm

```
node --version
npm --version
```

A suitable Node.js result is:

Expected Node.js result

v22.12.0 or newer

A Node 24 result may resemble:

Recommended example

v24.x.x

Do Not Continue Until Resolved

Do not continue if the terminal reports `command not found: node` or if the Node version is older than 22.12.0.

4 Phase 2: Protect the Repository with a Website Branch

4.1 Fetch the current repository

In GitHub Desktop:

1. Select the repository **CosmicUniversalismStatement**.
2. Confirm the current branch is `main`.
3. Click **Fetch origin**.
4. Allow GitHub Desktop to finish synchronizing.

4.2 Create the website branch

In GitHub Desktop:

1. Click **Current Branch**.
2. Click **New Branch**.
3. Enter the branch name:

Branch name`website`

4. Set `main` as the base branch.
5. Click **Create Branch**.
6. Click **Publish branch**.

Checkpoint

The top of GitHub Desktop must display:

Current Branch: `website`

Do not create or edit website files while GitHub Desktop displays `main`.

4.3 Verify the branch in Terminal

Open Terminal in the repository root and run:

Verify the active branch

```
git branch --show current
```

The exact result should be:

Successful Result

`website`

5 Phase 3: Configure PyCharm

5.1 Open the complete repository

In PyCharm:

1. Select **File** → **Open**.
2. Select the complete `CosmicUniversalismStatement` folder.
3. Do not select only `ResearchFiles/`.
4. Do not select only `cosmic_converter/`.
5. Allow PyCharm to finish indexing the project.

5.2 Install the Astro plugin

In PyCharm:

1. Open **PyCharm** → **Settings**.
2. Select **Plugins**.
3. Select **Marketplace**.
4. Search for Astro.
5. Install the official Astro plugin.
6. Restart PyCharm if requested.

Then inspect:

Astro language-server location

```
Settings
-> Languages & Frameworks
-> TypeScript
-> Astro
```

Enable Astro support if that option is available.

5.3 Configure Node.js in PyCharm

Open:

PyCharm Node configuration

```
Settings
-> Languages & Frameworks
-> Node.js
```

Confirm:

- Node interpreter: Node 22.12.0 or newer.
- Package manager: npm.

5.4 Open the built-in terminal

In PyCharm:

Open Terminal

```
View
-> Tool Windows
-> Terminal
```

Run:

Final PyCharm environment check

```
git branch --show current
node --version
npm --version
```

Record the results:

Check	Observed result
Active Git branch	
Node.js version	
npm version	
Astro plugin installed	Yes / No
Repository opened at root	Yes / No

6 Phase 4: Create the ChatGPT Project

6.1 Create the project

In ChatGPT Desktop:

1. Create a new Project.
2. Name it:

ChatGPT Project name

Cosmic Universalism Website

6.2 Add the permanent project instructions

Open the Project's instruction field and paste the following.

Copy-and-Paste Prompt: Project Instructions

```
PROJECT: Cosmic Universalism Website

REPOSITORY:
willmaddock/CosmicUniversalismStatement

LOCAL REPOSITORY:
CosmicUniversalismStatement

DEVELOPMENT BRANCH:
website

WEBSITE DIRECTORY:
website/
```

TECHNOLOGY:

Astro, TypeScript, HTML, modern CSS, Markdown, SVG, and lightweight browser JavaScript.

DEVELOPMENT APPLICATIONS:

- PyCharm for reviewing and editing source files
- GitHub Desktop for branches, commits, and pushes
- ChatGPT Work for planning, research organization, design, and content
- ChatGPT Codex for repository inspection, coding, commands, builds, and tests

PROTECTION RULES:

1. Never modify the main branch.
2. Confirm the current branch is website before making changes.
3. Keep new website source code inside website/.
4. Do not delete or rewrite existing repository files unless explicitly directed.
5. Do not modify the CU-Time converter mathematics without a separate technical review.
6. Do not commit node_modules/, dist/, .DS_Store, private keys, tokens, passwords, or secret .env files.
7. Make small, reviewable changes.
8. Explain planned file changes before implementing them.
9. Run npm run build after major changes.
10. Do not deploy until the local production build succeeds.
11. Do not commit, push, merge, or deploy unless explicitly asked.

CU CONTENT RULES:

1. Preserve sub-ctom and ctom as different terms.
2. Do not place the Observable Universe marker inside sub-ctom.
3. Keep scientific references separate from CU theoretical propositions.
4. Clearly label open problems, contradictions, and superseded interpretations.
5. Preserve the 3.108-trillion-year full Cosmic Breath framework.
6. Do not silently change established project terminology.

VISUAL DIRECTION:

A realistic combination of NASA scientific visualization, modern academic publishing, cinematic cosmic documentary, and restrained philosophical symbolism.

COLORS:

- Deep navy-black background
- White-gold for the Great Baking Will and sub-ztom
- Cyan for empirical references and the Observable Universe
- Translucent blue for compression structures
- Amber for open questions

WORKFLOW:

Plan in Work.

Implement and test in Codex.

Review files in PyCharm.

Review and commit changes in GitHub Desktop.

6.3 Add reference materials

Add these reference materials to the Project:

- The approved Cosmic Universalism homepage mockup.
- The website blueprint or outline.
- Any canonical research documents needed for the first pages.

Use recognizable filenames:

Suggested reference filenames

```
cosmic-universalism-homepage-reference.png  
cosmic-universalism-website-blueprint.md
```

Safety Rule

Do not upload passwords, GitHub tokens, private SSH keys, secret environment files, or unrelated personal documents.

7 Phase 5: Connect ChatGPT Desktop and PyCharm

7.1 Enable Work with Apps

In ChatGPT Desktop:

1. Open **ChatGPT** → **Settings**.
2. Find **Work with Apps**.
3. Enable the feature.
4. Open **Manage Apps**.
5. Confirm whether PyCharm appears as a supported application.
6. Grant only the permissions required for the project.

The exact interface wording can vary by desktop-app release.

7.2 Prepare the PyCharm window

Before attaching PyCharm:

- Keep only the Cosmic Universalism repository open.
- Close unrelated private projects.
- Close files containing credentials or secrets.
- Confirm the active branch is `website`.

7.3 Use reviewable changes

For the first implementation phases:

- Ask for a plan before edits.
- Ask for a list of files that will change.
- Review proposed diffs.
- Avoid automatic application of broad edits.
- Test each approved change locally.

8 Phase 6: Create the Initial Work Plan

Open ChatGPT Work inside the Cosmic Universalism Website Project.

Paste:

Copy-and-Paste Prompt: Phase 1 Planning

We are beginning the Cosmic Universalism website project.

Use the Project Instructions, website blueprint, and approved homepage reference image.

For this task, do not edit any files.

Create a concise Phase 1 implementation plan for a local Astro website. The first milestone must include only:

1. Astro project initialization inside website/
2. Global design tokens
3. Base page layout
4. Header navigation
5. Static homepage hero
6. Placeholder Cosmic Breath SVG
7. Three summary cards
8. Research classification legend

For each item, identify:

- the goal
- the expected folders and files
- dependencies
- accessibility requirements
- mobile behavior
- testing requirements
- completion criteria

Do not implement anything yet.

Do not modify the repository.

Do not commit, push, merge, or deploy.

Review the plan before proceeding.

Checkpoint

The plan should keep all new application files inside `website/`. It should not propose changing the converter mathematics or rewriting the research repository.

9 Phase 7: Inspect the Repository with Codex

Switch to ChatGPT Codex and open the local repository.

Paste:

Copy-and-Paste Prompt: Repository Inspection

Inspect this repository without changing any files.

Confirm:

1. The repository root
2. The current Git branch
3. Whether a `website/` directory already exists
4. Whether `package.json` exists at the repository root
5. Whether Node.js and npm are available
6. The detected Node.js and npm versions
7. Whether there are uncommitted changes
8. Which GitHub Pages workflow files currently exist
9. Which branches those workflows trigger on
10. Whether any existing workflow could deploy from the `website` branch

Do not install packages.

Do not create files.

Do not modify files.

```
Do not switch branches.  
Do not commit.  
Do not push.  
Do not merge.  
Do not deploy.  
  
Report the findings first.
```

Do Not Continue Until Resolved

If Codex reports that the current branch is `main`, stop. Return to GitHub Desktop and switch to `website`.

10 Phase 8: Initialize Astro

10.1 Preferred Codex initialization prompt

After the inspection confirms the correct branch, paste:

Copy-and-Paste Prompt: Initialize Astro

```
Initialize a new Astro website inside the website/ directory.
```

Requirements:

- Use the minimal Astro template.
- Use strict TypeScript.
- Install dependencies.
- Use npm as the package manager.
- Do not initialize a nested Git repository.
- Do not modify files outside `website/`.
- Do not modify the existing GitHub Pages workflow.
- Do not deploy.
- Do not commit or push.
- After creation, run `npm run build`.
- Report every created file.
- Report all warnings or errors.
- Confirm whether a nested `website/.git` directory exists.

```
Stop after the initial successful Astro build.
```

10.2 Manual terminal alternative

To perform initialization manually from the repository root:

Launch the Astro setup wizard

```
npm create astro@latest website
```


Choose options equivalent to:

Astro wizard selections

```
Template: Minimal
Install dependencies: Yes
TypeScript: Strict
Initialize a new Git repository: No
```

The wizard wording may vary slightly.

10.3 Expected project structure

After initialization, PyCharm should show:

Expected Astro structure

```
website/
|-- public/
|-- src/
|   |-- pages/
|   |   |-- index.astro
|-- .gitignore
|-- astro.config.mjs
|-- package.json
|-- package-lock.json
|-- README.md
|-- tsconfig.json
```

Do Not Continue Until Resolved

There should not be a nested `website/.git/` directory. A nested Git repository would interfere with the parent repository's source control.

11 Phase 9: Run and Test the Initial Website

11.1 Start the development server

In PyCharm's terminal:

Run the local development server

```
cd website
npm run dev
```

Astro normally reports an address similar to:

Local development address

```
http://localhost:4321/
```

Open the address in Safari or Chrome.

To stop the server, press:

Stop the server

```
Control + C
```

11.2 Run the production build

From `website/`:

Build the production website

```
npm run build
```

A successful build creates:

Generated build directory

```
website/dist/
```

11.3 Preview the production build

Preview the production build

```
npm run preview
```

Successful Result

The initial milestone is successful when:

- The development server launches.
- The local page opens in a browser.
- `npm run build` completes without errors.
- `npm run preview` displays the production build.

12 Phase 10: Review and Commit the Initial Astro Project

12.1 Inspect the Git changes

Return to GitHub Desktop.

The changes should be concentrated inside:

Expected change location

```
website/
```

Do not commit:

- node_modules/
- dist/
- .DS_Store
- secret .env files
- access tokens
- passwords
- private keys

Confirm that website/.gitignore includes:

Minimum ignore entries

```
node_modules
dist
```

12.2 Create the first commit

Use this commit summary:

Commit message

```
Create initial Astro website
```

In GitHub Desktop:

1. Confirm the button says **Commit to website**.
2. Review every selected file.
3. Click **Commit to website**.
4. Click **Push origin**.

Do Not Continue Until Resolved

Do not click the commit button if it says **Commit to main**.

13 Phase 11: Establish the Website Architecture

Use ChatGPT Work to prepare the architecture before implementation.

Copy-and-Paste Prompt: Website Architecture Plan

Create the implementation specification for the initial Cosmic Universalism website architecture.

Do not edit files.

The application root is website/.

Plan the following structure:

```
website/  
  public/  
    diagrams/  
    images/  
    media/  
    tools/  
  src/  
    components/  
    content/  
      framework/  
      research/  
      glossary/  
      media/  
      archive/  
    layouts/  
    pages/  
      research/  
    styles/
```

Plan these initial components:

- Header.astro
- Hero.astro
- CosmicBreathDiagram.astro
- BreathSummary.astro
- ResearchLegend.astro
- Footer.astro

Plan these initial layouts:

- BaseLayout.astro
- ResearchLayout.astro

Plan these initial pages:

- index.astro
- framework.astro

- cosmic-breath.astro
- cu-time.astro
- research/index.astro
- media.astro
- about.astro

Plan these initial stylesheets:

- tokens.css
- global.css
- typography.css

For every file, explain:

- its responsibility
- what it imports
- what content it owns
- what content it must not duplicate
- accessibility requirements
- responsive behavior
- completion criteria

Do not implement the files.

Do not modify the repository.

Do not commit, push, merge, or deploy.

14 Phase 12: Implement the Design Foundation

14.1 Planning prompt

Copy-and-Paste Prompt: Design System Specification

Using the approved Cosmic Universalism mockup and Project Instructions, prepare a detailed design-system specification.

Do not edit files.

Define:

1. Color tokens
2. Typography tokens
3. Spacing scale
4. Maximum content width
5. Research reading width
6. Border and surface styles
7. Button styles
8. Focus states
9. Research classification styles
10. Reduced-motion behavior

11. Desktop breakpoints
12. Mobile breakpoints

Use these visual principles:

- deep navy-black background
- white-gold for the Great Baking Will and sub-ztom
- cyan for empirical references and the Observable Universe
- translucent blue for compression structures
- amber for open questions
- high readability
- restrained glow
- no video-game HUD styling
- no cryptocurrency styling
- no excessive particle effects

Return the specification only.
Do not implement it.

14.2 Implementation prompt

After approving the specification, switch to Codex and paste:

Copy-and-Paste Prompt: Implement Design Tokens and Base Layout

Implement only the approved design tokens and base page layout.

Requirements:

- Work only inside website/.
- Create maintainable CSS custom properties.
- Create a semantic BaseLayout.astro.
- Add accessible metadata defaults.
- Add a skip-to-content link.
- Add visible keyboard focus styles.
- Add reduced-motion support.
- Do not build the final header or hero yet.
- Do not add heavy dependencies.
- Do not modify existing research or converter files.
- Run `npm run build`.
- Report all changed files.
- Report warnings and errors.
- Do not commit, push, merge, or deploy.

Suggested commit message after review:

Commit message

Add global CU design system and base layout

15 Phase 13: Build the Header and Static Hero

15.1 Work planning prompt

Copy-and-Paste Prompt: Header and Hero Specification

Using the approved homepage mockup, create an implementation specification for the homepage header and static hero only.

Do not edit files.

Specify:

- exact component names
- semantic HTML structure
- navigation labels
- accessible mobile navigation behavior
- hero heading hierarchy
- exact hero text
- button labels
- provisional button destinations
- responsive desktop and mobile layout
- placeholder behavior for the Cosmic Breath SVG
- keyboard focus behavior
- reduced-motion requirements
- acceptance criteria

Required navigation labels:

Framework
Cosmic Breath
CU-Time
Research
Media
About
GitHub

Required site identity:

COSMIC UNIVERSALISM
Computational Intelligence Initiative

Required hero heading:

COSMIC UNIVERSALISM

Required hero description:

A philosophical and computational framework exploring
consciousness, time, existence, and the full Cosmic Breath.

Required buttons:

Explore the Framework
Open the CU Statement

Required research-status line:

Scientific references distinguished from CU theoretical propositions.

Do not implement the specification.

15.2 Codex implementation prompt

Copy-and-Paste Prompt: Implement Header and Static Hero

Implement only the approved header and static homepage hero.

Requirements:

- Work only inside website/.
- Use semantic and accessible HTML.
- Use the existing design tokens.
- Keep content separate from decorative styling.
- Implement a responsive desktop and mobile arrangement.
- Use a static placeholder for the Cosmic Breath diagram.
- Do not add animation yet.
- Do not add a JavaScript framework.
- Do not modify research files.
- Do not modify converter mathematics.
- Run `npm run build`.
- Provide a file-by-file change summary.
- Report warnings and errors.
- Do not commit, push, merge, or deploy.

Suggested commit message:

Commit message

Add homepage header and static hero

16 Phase 14: Build the Cosmic Breath SVG

16.1 Planning prompt

Copy-and-Paste Prompt: Cosmic Breath SVG Specification

Create a technical specification for a responsive, accessible, static SVG visualization of the Cosmic Breath.

Do not edit files.

The SVG must include these exact labels:

- sub-ztom
- atom
- btom
- ctom
- ztom

The center must contain a white-gold sub-ztom seed.

The Observable Universe marker must:

- be cyan
- appear inside the ctom region
- display "You Are Here"
- display "Observable Universe"
- not appear inside sub-ctom
- not overlap the central seed

The diagram should look like a realistic scientific website visualization built from SVG shapes, gradients, strokes, masks, and labels.

It must not resemble:

- a gaming HUD
- a magical portal
- a cryptocurrency graphic
- a crowded science-fiction dashboard

Specify:

- SVG viewBox
- grouping strategy
- shell geometry
- text-label strategy
- responsive scaling
- accessibility title and description
- mobile fallback
- reduced-motion behavior for later animation
- acceptance criteria

Do not implement the SVG yet.

16.2 Implementation prompt

Copy-and-Paste Prompt: Implement Static Cosmic Breath SVG

Implement the approved static Cosmic Breath SVG.

Requirements:

- Work only inside website/.
- Use an Astro component named CosmicBreathDiagram.astro.
- Preserve the exact approved tom labels.
- Place the cyan Observable Universe marker inside ctom.
- Do not introduce a sub-ctom label unless the specification explicitly requires it.
- Add an accessible SVG title and description.
- Ensure text remains readable at supported desktop widths.
- Provide a simplified readable mobile arrangement.
- Do not animate the SVG yet.
- Run `npm run build`.
- Report changed files, warnings, and errors.
- Do not commit, push, merge, or deploy.

Suggested commit message:

Commit message

Add static Cosmic Breath SVG visualization

17 Phase 15: Build the Summary Cards and Research Legend

17.1 Required card content

Expansion card

Expansion Phase
2.8 trillion years
sub-ztom -> atom

Compression card

Compression Phase
308 billion years
btom -> ztom

Full Breath card

Full Cosmic Breath

3.108 trillion years
Expansion + Compression

17.2 Required research classifications

Research legend

Empirical Reference
CU Mathematical Model
CU Theoretical Proposition
Open Question

17.3 Implementation prompt

Copy-and-Paste Prompt: Implement Summary Cards and Research Legend

Implement the approved Cosmic Breath summary cards and research classification legend.

Requirements:

- Work only inside website/.
- Create BreathSummary.astro.
- Create ResearchLegend.astro.
- Use semantic HTML.
- Use the approved design tokens.
- Stack cards vertically on narrow screens.
- Preserve all numerical values and labels exactly.
- Use cyan for Empirical Reference.
- Use white-gold for CU Mathematical Model.
- Use translucent blue for CU Theoretical Proposition.
- Use amber for Open Question.
- Do not add unsupported scientific claims.
- Do not add animation.
- Run `npm run build`.
- Provide a file-by-file change summary.
- Do not commit, push, merge, or deploy.

Suggested commit message:

Commit message

Add Cosmic Breath summary and research legend

18 Phase 16: Repeatable Feature Workflow

Use the following sequence for every feature:

1. Plan one small feature in ChatGPT Work.
2. Review the plan.
3. Implement only that feature in Codex.
4. Review every changed file in PyCharm.
5. Run the development server.
6. Inspect the feature in the browser.
7. Test desktop and mobile widths.
8. Run the production build.
9. Review GitHub Desktop changes.
10. Commit to website.
11. Push origin.

18.1 Generic planning prompt

Copy-and-Paste Prompt: Generic Feature Planning

Plan the following website feature:

[INSERT FEATURE NAME]

Do not edit files.

Repository rules:

- Current development branch: website
- Website application directory: website/
- Do not modify main
- Do not modify existing research or converter files unless explicitly directed
- Do not change CU-Time mathematics
- Preserve CU terminology exactly

Provide:

1. Goal
2. User experience
3. Files to create
4. Files to modify
5. Semantic HTML structure
6. Accessibility requirements
7. Responsive behavior

8. Data or content requirements
9. Testing steps
10. Completion criteria
11. Risks or unresolved questions

Do not implement.

Do not commit, push, merge, or deploy.

18.2 Generic implementation prompt

Copy-and-Paste Prompt: Generic Feature Implementation

Implement only this approved feature:

[INSERT FEATURE NAME]

Approved specification:

[PASTE OR REFERENCE THE APPROVED SPECIFICATION]

Rules:

- Confirm the current branch is website before editing.
- Work only inside website/ unless the approved specification explicitly identifies another file.
- Make the smallest maintainable change.
- Use semantic HTML.
- Preserve accessibility.
- Preserve responsive behavior.
- Do not add unnecessary dependencies.
- Do not modify CU-Time mathematics.
- Do not alter research conclusions.
- Run `npm run build` after editing.
- Report every changed file.
- Report all warnings and errors.
- Do not commit, push, merge, or deploy.

18.3 Generic review prompt

Copy-and-Paste Prompt: Review the Completed Feature

Review the current uncommitted website changes.

Do not edit files during the first review.

Check:

1. Whether all changes are on the website branch
2. Whether files outside website/ changed

3. Whether the implementation matches the approved specification
4. Accessibility
5. Responsive behavior
6. Broken links
7. Incorrect paths
8. Incorrect CU terminology
9. sub-ctom versus ctom confusion
10. Observable Universe marker placement
11. Build warnings
12. Unnecessary dependencies
13. Secret or generated files that should not be committed
14. Whether npm run build succeeds

Report findings from highest to lowest priority.
Do not commit, push, merge, or deploy.

19 Phase 17: Recommended Feature Order

Develop features in this order:

1. Initial Astro application
2. Global design tokens
3. Base layout
4. Header
5. Static hero
6. Static Cosmic Breath SVG
7. Summary cards
8. Research-classification legend
9. Mobile navigation
10. Footer
11. Framework overview page
12. CU Statement page
13. Cosmic Breath page
14. Research index
15. Research article layout

16. Glossary
17. CU-Time converter wrapper
18. Media page
19. Subtle motion
20. Reduced-motion review
21. Accessibility review
22. Production deployment

Safety Rule

Do not begin complex animation, WebGL, search, a database, a Go backend, or user accounts during the initial website milestone.

20 Phase 18: Local Quality Checks

Run these commands from `website/`:

Development server

```
npm run dev
```

Production build

```
npm run build
```

Production preview

```
npm run preview
```

Use this checklist:

- ☐ Homepage loads.
- ☐ No browser-console errors appear.
- ☐ All navigation items are visible.
- ☐ Keyboard focus is visible.
- ☐ The skip link works.
- ☐ The hero remains readable.

- ☐ The SVG labels are correct.
- ☐ The Observable Universe marker is inside ctom.
- ☐ Summary values are correct.
- ☐ Research classifications are distinguishable.
- ☐ The page works at mobile width.
- ☐ The page works at desktop width.
- ☐ Reduced-motion settings are respected.
- ☐ `npm run build` succeeds.

21 Phase 19: Commit Discipline

Use small, descriptive commits.

Recommended sequence:

Suggested commit history

```
Create initial Astro website
Add global CU design system and base layout
Add homepage header and static hero
Add static Cosmic Breath SVG visualization
Add Cosmic Breath summary and research legend
Add responsive mobile navigation
Add Framework overview page
Add CU Statement page
Add research library structure
Integrate CU-Time converter page
Add reduced-motion support
Prepare website branch deployment
```

Before every commit:

- ☐ GitHub Desktop displays website.
- ☐ The changed-file list has been reviewed.
- ☐ No secret files are selected.
- ☐ No `node_modules/` files are selected.
- ☐ No `dist/` files are selected.
- ☐ The production build succeeds.
- ☐ The commit message describes one logical change.

22 Phase 20: GitHub Pages Deployment

Do Not Continue Until Resolved

Do not perform this phase until the local production build has been reviewed and approved. The repository may already contain a GitHub Pages workflow. Inspect all existing workflows before adding another deployment workflow. If an existing workflow deploys from main, future pushes to main could overwrite a site deployed from website.

22.1 Configure Astro for the repository path

Edit website/astro.config.mjs:

astro.config.mjs

```
import { defineConfig } from 'astro/config';

export default defineConfig({
  site: 'https://willmaddock.github.io',
  base: '/CosmicUniversalismStatement',
  output: 'static',
});
```

Because the site uses a repository subpath, manually written internal links must account for the configured base path.

22.2 Deployment-review prompt

Before creating the workflow, use:

Copy-and-Paste Prompt: Deployment Safety Review

```
Inspect the repository's current GitHub Actions workflows.

Do not edit files.

Determine:

1. Which workflows deploy GitHub Pages
2. Which branches trigger each workflow
3. Whether a main-branch push could overwrite a deployment from
   the website branch
4. Whether GitHub Pages is currently configured for Actions
5. Whether the Astro project path should be ./website
6. Whether astro.config.mjs has the correct site and base values
7. Whether package-lock.json is committed
8. Whether all internal links support the configured base path

Recommend the safest deployment arrangement for keeping:
```

- main as the canonical research branch
- website as the permanent website branch

Do not modify workflows.

Do not commit, push, merge, or deploy.

22.3 Website-branch deployment workflow

As of the verification date shown on this document, the official Astro documentation uses:

- actions/checkout@v7
- withastro/action@v6
- actions/deploy-pages@v5

Create this file only after reviewing existing workflows:

.github/workflows/deploy-website.yml

```
name: Deploy Cosmic Universalism Website

on:
  push:
    branches: [website]
  workflow_dispatch:

permissions:
  contents: read
  pages: write
  id-token: write

concurrency:
  group: pages
  cancel-in-progress: false

jobs:
  build:
    runs-on: ubuntu-latest

    steps:
      - name: Checkout repository
        uses: actions/checkout@v7

      - name: Install, build, and upload Astro site
        uses: withastro/action@v6
        with:
          path: ./website
          node-version: 24

  deploy:
```

```
needs: build
runs-on: ubuntu-latest

environment:
  name: github-pages
  url: ${ steps.deployment.outputs.page_url }}

steps:
  - name: Deploy to GitHub Pages
    id: deployment
    uses: actions/deploy-pages@v5
```

GitHub Action major versions can change. Recheck the official Astro GitHub Pages documentation before committing this workflow.

22.4 Configure GitHub Pages

On GitHub:

GitHub Pages settings

Repository
-> Settings
-> Pages
-> Source
-> GitHub Actions

The expected public address is:

Expected public URL

<https://willmaddock.github.io/CosmicUniversalismStatement/>

22.5 Deployment implementation prompt

Copy-and-Paste Prompt: Prepare Deployment

Prepare the approved GitHub Pages deployment for the Astro website.

Requirements:

- Confirm the current branch is website.
- Inspect existing workflow files before editing.
- Do not allow an old workflow to unexpectedly overwrite the website deployment.
- Configure website/astro.config.mjs with:
site: <https://willmaddock.github.io>
base: /CosmicUniversalismStatement

```
output: static
- Create or update only the approved website deployment workflow.
- Use ./website as the Astro project path.
- Run npm run build locally.
- Check internal links and asset paths against the base path.
- Report every changed file.
- Do not commit or push.
- Do not trigger a deployment.
```

23 Phase 21: Troubleshooting

23.1 Node command not found

Symptom:

Error

```
zsh: command not found: node
```

Response:

- Install a supported Node.js release.
- Close and reopen Terminal.
- Run `node -version`.

23.2 Wrong branch

Symptom:

Unexpected result

```
main
```

Response:

1. Stop editing.
2. Save or discard uncommitted changes appropriately.
3. Open GitHub Desktop.
4. Switch to `website`.
5. Run `git branch -show-current` again.

23.3 Nested Git repository

Symptom:

Unwanted directory

```
website/.git/
```

Response:

- Do not commit yet.
- Remove only the nested `website/.git/` directory.
- Do not remove the parent repository's `.git/`.
- Recheck GitHub Desktop.

23.4 Port 4321 is already in use

Run Astro on another port:

Alternative local port

```
npm run dev -- --port 4322
```

Then open:

Alternative address

```
http://localhost:4322/
```

23.5 PyCharm does not recognize Astro files

Check:

- The Astro plugin is installed and enabled.
- PyCharm was restarted after installation.
- The complete repository is open.
- Node.js is configured.
- Astro language-server support is enabled.

23.6 GitHub Pages returns a 404

Check:

- The workflow succeeded.

- Pages source is set to GitHub Actions.
- `site` is correct.
- `base` is `/CosmicUniversalismStatement`.
- Internal links include the repository base path.
- Static asset paths are compatible with the base path.

23.7 The build fails

Run:

Clean dependency installation

```
cd website
rm -rf node_modules
npm install
npm run build
```

Do not delete `package-lock.json` unless dependency recovery specifically requires regenerating it.

24 Phase 22: Initial Project Checkpoint

Complete only these tasks before beginning homepage implementation:

- ☐ Fetch origin in GitHub Desktop.
- ☐ Create the `website` branch.
- ☐ Publish the `website` branch.
- ☐ Verify the branch in Terminal.
- ☐ Verify Node.js.
- ☐ Verify npm.
- ☐ Open the repository root in PyCharm.
- ☐ Install the Astro plugin.
- ☐ Create the ChatGPT Project.
- ☐ Add the Project Instructions.
- ☐ Add the homepage reference image.
- ☐ Add the website blueprint.
- ☐ Run the planning prompt.

- ☐ Run the no-edit repository inspection.

Record the checkpoint:

Checkpoint item	Result
Current branch	
Node.js version	
npm version	
PyCharm Astro plugin	
ChatGPT Project created	
Reference mockup added	
Blueprint added	
Repository inspection complete	
Uncommitted changes before setup	
Existing deployment workflows	

25 Official Reference Links

These links were used to establish the technical baseline of this manual.

- Astro installation and Node.js requirements:
<https://docs.astro.build/en/install-and-setup/>
- Astro editor and JetBrains setup:
<https://docs.astro.build/en/editor-setup/>
- Astro GitHub Pages deployment:
<https://docs.astro.build/en/guides/deploy/github/>
- GitHub Desktop branch management:
[GitHub Desktop branch-management documentation](#)
- ChatGPT Projects:
[OpenAI Projects documentation](#)
- Work with Apps on macOS:
[OpenAI Work with Apps documentation](#)
- ChatGPT Work and Codex:
[OpenAI Work and Codex documentation](#)

Final Operating Principle

Controlled Development Rule

Plan one feature.
Implement one feature.
Review every changed file.
Test locally.
Run the production build.
Commit only to `website`.
Push only after the change is understood.
Do not deploy until the local website is stable.